Российская экономическая школа

Негосударственное образовательное учреждение высшего образования

Российская экономическая школа (институт) (РЭШ)

Программа по направлению подготовки
38.04.01 Экономика (уровень магистратуры)
направленность (профиль) «Магистр экономики»

ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА

MASTER THESIS

Тема: Грубая Волатильность на Финансовых Рынках: Моделирование и Прогнозирование Реализованной Волатильности

Title: Rough Volatility in Financial Markets: Modeling and Forecasting Realized Volatility

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Оценка/Grade

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Москва 2026

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Abstract

This paper revisits the multivariate fractional Brownian motion (mfBm) framework of Bibinger, Yu and Zhang (2025) for forecasting log-realized variance, applying it to a panel of MOEX equities. We introduce three methodological refinements. First, all comparisons are conducted under the proxy-robust QLIKE loss with formal Diebold–Mariano tests. Second, heterogeneity of mfBm forecasting performance is explored through the asset's Hurst exponent. Third, the companion assets that enter the multivariate forecast are selected dynamically, which employ the Bibinger et al.(2025) theoretical optimality condition.

Under QLIKE, the univariate fBm beats HAR at long horizons by a modest but cross-sectionally consistent margin, while HAR dominates at shorter horizons; reporting under MSFE materially inflates the apparent size of the fractional advantage. The fractional advantage is sharply roughness-conditional: the regression of the one-day QLIKE gain on $\bar{H}$ exhibits a strongly negative slope with substantial cross-sectional explanatory power, so that the fractional framework offers little improvement on assets whose log-volatility is closer to the long-memory regime. The dynamic-bundle mfBm produces a systematic but small cross-sectional improvement over the univariate fBm at intermediate and long horizons, growing with the bundle size. For the overwhelming majority of MOEX anchors at any bundle size and horizon, however, the per-anchor DM test cannot reject equality of mfBm and fBm: the case for the multivariate model is cross-sectional rather than per-asset.

1 Introduction and Literature Review

1.1 What Is Volatility and Why We Care About It

Volatility is the central input to risk management, option pricing, and portfolio construction — which makes forecasting it a persistently important problem. In each of these domains the relevant quantity is forward-looking: the value-at-risk of a trading book depends on the distribution of losses the institution may face over the next trading day; an investor seeking to construct an efficient portfolio cares about the covariation structure that will prevail over her investment horizon, not the one that prevailed in the past.

1.2 Empirical Properties of Volatility and the Persistence Debate

The stylized fact that motivates this paper is the slow decay of the autocorrelation function of log-realized variance: autocorrelations remain positive and significant out to lags of several months (Andersen et al., 2003; Corsi, 2009), far beyond what short-memory ARMA models can reproduce. The traditional reading is that volatility exhibits *long memory*, with autocorrelations decaying at a hyperbolic rather than exponential rate.

A recent literature challenges this reading. Gatheral et al. (2018) show that the small-lag scaling of the absolute increments of log-volatility yields a scaling exponent well below 1/2, implying that paths are locally rougher than Brownian motion and that increments are *negatively* autocorrelated at short horizons — the opposite of what long memory implies. The two views refer to different time scales (small-lag local versus large-lag global), and Gatheral et al. (2018) show that the apparent long-memory autocorrelation signature is itself consistent with a stationary rough process. Which reading provides the more useful description for forecasting purposes is an open empirical question.

1.3 From Antipersistence to Roughness

The anti-persistent reading of log-volatility calls for a modelling framework in which local irregularity is a primitive. The natural candidate is the *fractional Brownian motion* (fBm), in which a single Hurst exponent $H \in (0,1)$ governs both the local regularity of sample paths and the long-run correlation structure of increments: $H = 1/2$ recovers standard Brownian motion, $H > 1/2$ yields positively correlated increments and smoother paths, and $H < 1/2$ yields negatively correlated increments and locally rougher paths.

Gatheral et al. (2018) argue that log-realized variance for major equity indices is well described by an fBm with $H \approx 0.1$, far below the smooth-rough threshold. Fukasawa et al. (2019) confirm $\hat{H} \ll 1/2$ on S&P 500 data with a quasi-likelihood estimator that explicitly accounts for the estimation error in realized variance. The reading has not gone uncontested: Cont and Das (2023) argue that the small scaling exponents may be a statistical artefact of how realized variance is constructed, rather than a property of the latent spot variance. Dandapani et al. (2021), in the other direction, provide a microstructural foundation: nearly unstable quadratic Hawkes processes — a natural model for endogenous order flow — converge in their scaling limit to a rough stochastic volatility model with $H < 1/2$ governed by the tail decay of the self-excitation kernel.

Here we do not attempt to resolve this debate, but rather adopt pragmatic view as follows. We take the realized-variance series as the observable object of interest and ask whether modelling its dynamics as a fractional process produces useful out-of-sample forecasts relative to HAR. The univariate fBm provides the baseline and the multivariate generalization of Bibinger et al. (2025) allows joint modelling of several realized-variance series with possibly heterogeneous roughness, which is empirically attractive given the cross-sectional variability of Hurst estimates across MOEX assets.

1.4 Contributions of This Paper

Starting point for this paper is a Bibinger et al. (2025), hereafter BYZ. BYZ generalizes the univariate fractional Brownian motion model of log-realized variance to a multivariate setting in which each component have its own Hurst exponent and the components interact by a cross-correlation matrix. The paper's central theoretical insight is a sharp characterization of when the multivariate specification strictly improves on the univariate one: gains arise from the simultaneous presence of heterogeneity in the Hurst exponents and non-zero cross-correlation. The theoretical result is substantial, but the accompanying empirical exercise in BYZ leaves several open questions. Forecast performance in BYZ is reported under the mean-squared forecast error (MSFE), which Patton (2011) shows is not robust to the use of an imperfect realized-variance proxy and can therefore deliver misleading rankings of competing forecasts. Also, no formal inference is conducted on the observed forecast-error differentials. The question of when the fractional framework improves on the heterogeneous-autoregressive benchmark of Corsi (2009), *conditional* on the very roughness level that the framework is built around, is not addressed. Also, companion assets entering the multivariate forecast are selected arbitrarily, a choice that bears no relation to the theoretical condition the paper itself derives.

We make four contributions, each paired with its empirical finding.

1. **Evaluation under a proxy-robust loss (QLIKE).** Every comparison in this paper is conducted under the QLIKE loss identified by Patton (2011) as robust to the realized-variance proxy, with MSFE retained only as a cross-check. The choice of metric matters materially for the timing of the fBm–HAR crossover: under MSFE the cross-sectional mean gain of fBm over HAR turns positive already at forecasting horizon $h \approx 6$, whereas under the proxy-robust QLIKE the crossover is delayed to $h \approx 15$. Reporting under MSFE alone, as in BYZ, therefore credits the fractional framework with an advantage at horizons at which it has none under the proxy-robust criterion.

2. **Formal inference on forecast-error difference.** Every pairwise model comparison is accompanied by the Diebold–Mariano test of Diebold and Mariano (1995) with a Bartlett HAC variance estimator. The inference reveals substantial cross-sectional heterogeneity: at $h = 1$ the test rejects in favour of HAR for 37% of the 43 MOEX assets and in favour of fBm for 7%; by $h = 10$ the HAR-wins fraction has fallen to 9%, and it stays around 7% through the rest of the competitive range.

3. **Roughness-conditional characterization of fBm versus HAR.** Before comparing mfBm to fBm, we need to know whether fBm beats HAR at all, and how that depends on the asset's roughness. A cross-sectional OLS regression of the QLIKE gain on the time-series mean Hurst estimate $\bar{H}_i$ gives a slope of -60.3 ($p < 0.001$, $R^2 = 0.43$) at the one-day horizon. The contrast between AKRN ($\bar{H} \approx 0.147$), for which the DM test rejects in favour of fBm at most horizons, and SBER ($\bar{H} \approx 0.278$), for which it rejects in favour of HAR at every short horizon, captures the two extremes of the cross-section. The fractional framework is worth deploying *conditional* on the asset being sufficiently rough; on assets whose log-volatility is closer to the long-memory regime, the HAR benchmark remains hard to beat at short horizons.

4. **Theory-consistent dynamic selection of companion assets.** We propose replacing BYZ's arbitrary selection rule with a dynamic, theory-consistent one: at every rolling forecast date the $d-1$ companions are chosen to maximize the score $|\hat{H}_{i,t} - \hat{H}_{j,t}| \cdot |\hat{\rho}_{i,j,t}|$, which employs BYZ's optimality condition. Empirically, the dynamic-bundle mfBm produces a *systematic but small* cross-sectional improvement over the univariate fBm at $h \geq 5$: the cross-sectional mean MSFE gain of mfBm$_d$ over fBm rises from +0.12% at $d = 1$ to +0.42% at $d = 9$ at $h = 14$, and at $h = 10$ roughly seventy to eighty percent of MOEX anchors have a lower sample MSFE under mfBm. The improvement, however, is rarely large enough to register on per-anchor Diebold–Mariano tests: between 86% and 100% of anchors at any (d, h) are statistically indistinguishable from the univariate fBm at $\alpha = 0.05$. The case for the multivariate model is therefore cross-sectional rather than per-anchor. As a separate finding, the BYZ theoretical monotonicity ($\text{MSFE}_{\text{mfBm}} \leq \text{MSFE}_{\text{fBm}}$ under correct specification and

known parameters) fails at short horizons with large bundles: at $(h, d) = (1, 9)$, 72% of anchors are worse under mfBm in sample MSFE and up to 30% are DM-significantly worse under MSFE.

A fifth, complementary dimension of the empirical work concerns the market in which it is conducted. No prior study of multivariate rough volatility has, to our knowledge, been applied to Russian financial markets. The microstructure, liquidity, and regime characteristics of MOEX differ markedly from those of the US large-cap universes that have dominated the rough-volatility literature.

2 Theoretical Framework

Here we cover realized variance, the HAR benchmark, and the fractional Brownian motion models that drive the empirical work.

2.1 Realized Variance from High-Frequency Data

Let S_t denote the price of a financial asset and $X_t = \log S_t$ its logarithm. We assume that $X = \{X_t, t \geq 0\}$ is a continuous semimartingale with stochastic differential

$$\mathrm{d}X_t = \mu_t \, \mathrm{d}t + \sigma_t \, \mathrm{d}W_t, \tag{1}$$

where μ_t is the drift, $\sigma_t > 0$ is the càdlàg spot volatility, and W is a standard Brownian motion adapted to the relevant filtration. The *integrated variance* over the trading day $[t-1, t]$ is the random variable

$$\mathrm{IV}_t \stackrel{\text{def}}{=} \int_{t-1}^{t} \sigma_u^2 \, \mathrm{d}u, \tag{2}$$

which is unobservable but represents the relevant measure of total within-day variation for pricing and risk applications.

Suppose now that within day t we observe M intraday log-prices at the times $t - 1 = \tau_0 < \tau_1 < \cdots < \tau_M = t$, with corresponding log-returns $r_{t,i} = X_{\tau_i} - X_{\tau_{i-1}}$. The *realized variance* of day t is defined as the sum of squared intraday log-returns,

$$\mathrm{RV}_t \stackrel{\text{def}}{=} \sum_{i=1}^{M} r_{t,i}^2. \tag{3}$$

Under standard regularity conditions on the model (1), Andersen et al. (2003) show that the realized variance is an asymptotically unbiased and consistent estimator of the integrated variance:

$$\mathrm{RV}_t \xrightarrow{\mathbb{P}} \mathrm{IV}_t \quad \text{as} \quad \max_{1 \leq i \leq M} (\tau_i - \tau_{i-1}) \to 0. \tag{4}$$

In practice the sampling frequency cannot be made arbitrarily fine without contaminating the estimator with bid–ask bounce, discrete price grids, and other forms of high-frequency noise that violate the underlying semimartingale assumption. At sub-minute frequencies, the bias from microstructure noise dominates, and the realized variance estimator can diverge from the true integrated variance (Andersen et al., 2003). A standard approach is to sample at the five-minute frequency, which empirically balances the bias induced by microstructure noise against the variance reduction from finer sampling.

Throughout the paper the object of empirical interest is the *log-realized variance*, $y_t \stackrel{\text{def}}{=} \log \mathrm{RV}_t$. Working in logs is motivated by two facts: y_t is empirically much closer to Gaussianity than RV_t, and the rough-volatility literature has settled on log-volatility as the process whose scaling and fractional properties are to be modeled.

2.2 The Heterogeneous Autoregressive Model

The benchmark model for realized-variance forecasting in the applied literature is the Heterogeneous Autoregressive (HAR) model of Corsi (2009). The HAR model approximates the long-memory dynamics of log-realized variance through a parsimonious linear regression on three lagged realized-volatility components at different time scales. For a forecast horizon $h \geq 1$, the direct h-step HAR specification is

$$y_{t+h} = \beta_0 + \beta_d\, y_t + \beta_w\, y_t^{(w)} + \beta_m\, y_t^{(m)} + \varepsilon_{t+h}, \tag{5}$$

where $y_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} y_{t-k}$ and $y_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} y_{t-k}$ denote the simple averages of log-realized variance over the previous trading week and month, respectively. Parameters are estimated by ordinary least squares separately for each forecast horizon.

Appeal of this model lies in simplicity, robustness, and reliable out-of-sample performance: the model has proved difficult to beat in the volatility forecasting literature, even when more elaborate parametric specifications are used. Reported out-of-sample Mincer–Zarnowitz R^2 statistics for HAR on equity-index log-realized variance are typically of the order of 0.4–0.6 at the one-day horizon. We will use HAR throughout the empirical part of the paper as the benchmark against which the fractional models are evaluated.

2.3 Roughness and Hölder Continuity

To formalize the notion of roughness introduced informally above, we recall the concept of Hölder continuity. A function $f : [0, T] \to \mathbb{R}$ is said to be *Hölder continuous of order* $\alpha \in (0, 1]$ on $[0, T]$ if there exists a constant $C > 0$ such that

$$|f(t) - f(s)| \leq C\,|t - s|^{\alpha} \qquad \text{for all } s, t \in [0, T]. \tag{6}$$

For a stochastic process $\{X_t\}$, the Hölder regularity of its sample paths quantifies their local irregularity: a higher Hölder exponent corresponds to smoother paths and a lower one to rougher paths. It can be seen from the definition that when t approaches s lower values of α result in left part of 6 being bounded by larger value i.e. lower α constitutes more erratic/rough behavior on short time scale. For the standard Brownian motion, almost every sample path is Hölder continuous of order α for every $\alpha < 1/2$. Processes with Hölder regularity below 1/2 are therefore in a precise mathematical sense *rougher* than Brownian motion. The Hurst exponent H, introduced in the next subsection, plays the role of a path regularity index: for the fBm with parameter H, almost every sample path is Hölder continuous of every order $\alpha < H$.

2.4 Fractional and Multivariate Fractional Brownian Motion

This subsection introduces the univariate and multivariate fractional Brownian motion models that form the basis of the empirical work. The subsection is entirely based on BYZ (Bibinger et al., 2025).

2.4.1 Formulation

A *fractional Brownian motion* with Hurst exponent $H \in (0,1)$ is a centered Gaussian process $\{B_t^H, t \geq 0\}$ with $B_0^H = 0$ and covariance function

$$\operatorname{Cov}\big(B_s^H,\, B_t^H\big) = \tfrac{1}{2}\big(\,|s|^{2H} + |t|^{2H} - |s-t|^{2H}\big), \qquad s,t \geq 0. \tag{7}$$

For $H = 1/2$, equation (7) reduces to $\min(s,t)$, the covariance of standard Brownian motion. For $H \neq 1/2$ the increments of B^H are no longer independent, and the process is no longer a semimartingale.

A *d-dimensional multivariate fractional Brownian motion* (mfBm) is a centered Gaussian process $\mathbf{B}_t^{\mathbf{H}} = (B_t^{H_1}, \ldots, B_t^{H_d})^\top$ with component-wise Hurst exponents $\mathbf{H} = (H_1, \ldots, H_d) \in (0,1)^d$ and a cross-component covariance structure that generalizes (7). In the *time-reversible* variant adopted throughout the paper (see Section 2.4.3 below), the cross-covariance between components i and j at times s and t is

$$\operatorname{Cov}\Big(B_s^{H_i},\, B_t^{H_j}\Big) = \rho_{ij}\,\sigma_i\,\sigma_j\,\tfrac{1}{2}\big(\,|s|^{H_i+H_j} + |t|^{H_i+H_j} - |s-t|^{H_i+H_j}\big)\,, \tag{8}$$

where $\sigma_i > 0$ is a component-specific scale parameter and $\rho_{ij} \in [-1,1]$ is a cross-correlation parameter.

2.4.2 Basic Properties

The univariate fBm B^H enjoys two distinguishing properties. It is *self-similar* of order H, meaning that for any $a > 0$ the rescaled process $\{a^{-H} B_{at}^H\}$ is distributed identically to $\{B_t^H\}$. And its increments are *stationary*: the law of $B_{t+s}^H - B_t^H$ depends only on s, not on t. For $H = 1/2$ these properties reduce to the familiar self-similarity and independent-increment properties of Brownian motion. For $H \neq 1/2$ the increments are stationary but *dependent*: they are positively correlated for $H > 1/2$ (long memory) and negatively correlated for $H < 1/2$ (anti-persistence or roughness). This signed dependence is made precise by the autocovariance of the increment process. For discrete observations at intervals $\Delta > 0$, define the lag-1 increment $\Delta_k B^H = B_{k\Delta}^H - B_{(k-1)\Delta}^H$. Applying (7), the autocovariance of the resulting *fractional Gaussian noise* (fGN) at integer lag $\ell = i - j \geq 0$ is (Bibinger et al., 2025, Eq. 2)

$$\operatorname{Cov}\big(\Delta_i B^H,\, \Delta_j B^H\big) = \frac{\Delta^{2H}}{2}\big((\ell+1)^{2H} + (\ell-1)^{2H} - 2\ell^{2H}\big). \tag{9}$$

Setting $\ell = 1$ yields $\Delta^{2H}(2^{2H-1} - 1)$, which is strictly negative for $H < 1/2$ and strictly positive for $H > 1/2$. The following simulation plot shows difference of behavior of mfBm with different H values.

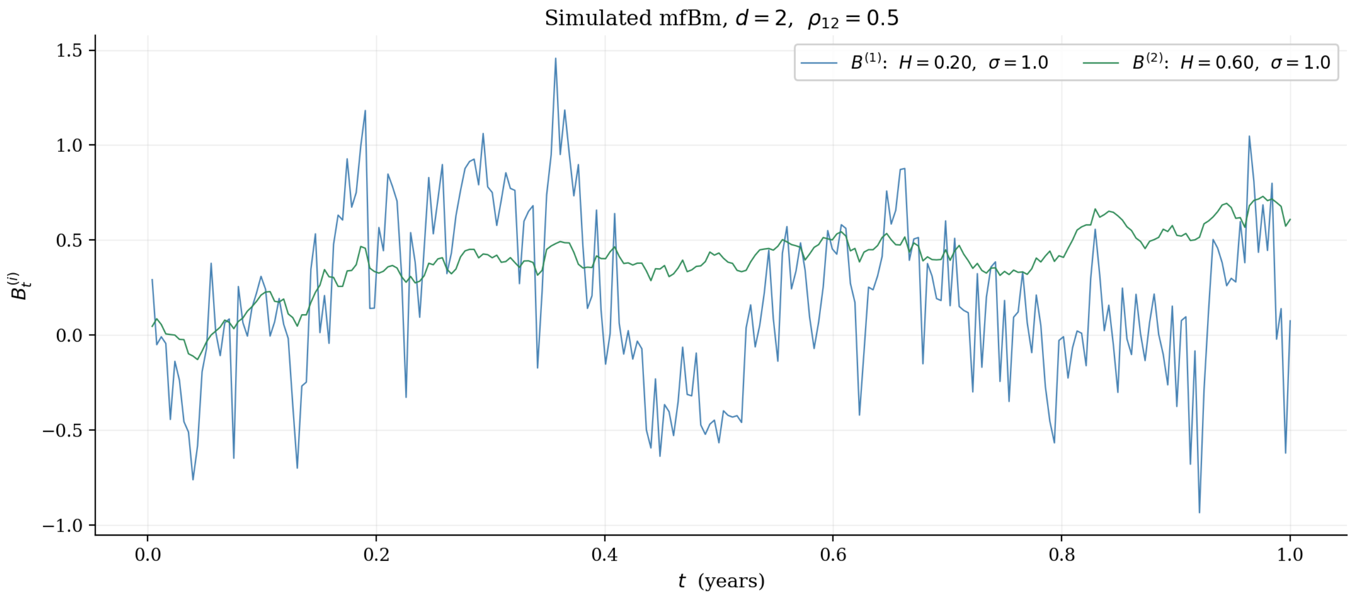


Figure 1: Simulated bivariate mfBm path on the unit time interval with heterogeneous Hurst exponents $(H_1, H_2) = (0.20, 0.60)$, common scale $\sigma_i = 1.0$, and cross-correlation $\rho_{12} = 0.5$. The first component $(H = 0.20)$ is locally very irregular; the second $(H = 0.60)$ is smoother and exhibits visible persistence.

2.4.3 Symmetric versus General Covariance and Time-Reversibility

In general, the cross-covariance of a d-dimensional mfBm *need not be symmetric* in the time arguments: one can have $\mathrm{Cov}(B_s^{H_i}, B_t^{H_j}) \neq \mathrm{Cov}(B_t^{H_i}, B_s^{H_j})$ for $i \neq j$. This asymmetry permits asymmetric lead–lag relationships between components, and the general specification includes an additional parameter governing the degree of asymmetry. The corresponding process is, however, not *time-reversible*: the joint distribution of $\mathbf{B}^{\mathbf{H}}$ run forward in time differs from the joint distribution of the time-reversed process.

For the financial applications considered in this paper, the symmetric, time-reversible variant in (8) is empirically preferred: the BYZ time-reversibility test is typically not rejected on realized-volatility data, the symmetric specification has fewer parameters to identify in moderate samples, and the optimal forecasting machinery of BYZ is derived under this variant. We accordingly restrict attention to the time-reversible mfBm throughout, and refer to the model defined by (8) when speaking of the multivariate fBm.

2.4.4 Parameter Estimation

One of the theoretical contributions of BYZ is the derivation of a novel moment-based estimator for the mfBm parameter vector $(\mathbf{H}, \boldsymbol{\sigma}, \boldsymbol{\rho})$. Hurst exponents and scale parameters are estimated component-wise from the marginal distributions, while cross-correlations are estimated pairwise. Following the notation of

BYZ, define the lag-1 and lag-2 increments for component j as

$$\Delta_k B^{(j)} = B^{(j)}_{k\Delta} - B^{(j)}_{(k-1)\Delta}, \quad k = 1, \ldots, n,$$
$$\Delta_{k,2} B^{(j)} = B^{(j)}_{(k+1)\Delta} - B^{(j)}_{(k-1)\Delta}, \quad k = 1, \ldots, n-1.$$

The Hurst exponent is estimated component-wise by (BYZ Eq. 6)

$$\hat{H}_j = \frac{1}{2\log 2} \log\left(\frac{\sum_{k=1}^{n-1} \left(\Delta_{k,2} B^{(j)}\right)^2}{\sum_{k=1}^{n} \left(\Delta_k B^{(j)}\right)^2} \right), \quad j = 1, \ldots, d. \tag{10}$$

The scale parameter is estimated component-wise by (BYZ Eq. 7)

$$\hat{\sigma}_j^2 = \frac{\sum_{k=1}^{n} \left(\Delta_k B^{(j)}\right)^2}{n\,\Delta^{2\hat{H}_j}}, \quad j = 1, \ldots, d, \tag{11}$$

and the cross-correlation between components 1 and 2 is estimated pairwise by (BYZ Eq. 8a)

$$\hat{\rho}_{12} = \frac{\sum_{k=1}^{n} \Delta_k B^{(1)} \cdot \Delta_k B^{(2)}}{\sqrt{\sum_{k=1}^{n} \left(\Delta_k B^{(1)}\right)^2 \cdot \sum_{k=1}^{n} \left(\Delta_k B^{(2)}\right)^2}}, \tag{12}$$

which is simply the Pearson correlation of the lag-1 increments. BYZ establish that all three estimators are *consistent* and *asymptotically normal* under mild regularity conditions.

2.4.5 Optimal Forecasting

Because $\mathbf{B^H}$ is jointly Gaussian with zero mean, the optimal forecast — the one that minimises the MSFE — coincides with the conditional expectation and is computable in closed form. Following BYZ (Section 4), consider t equispaced discrete observations stacked as a column vector

$$\mathcal{X}_{t,d} = \left(B^{(1)}_{\Delta}, \ldots, B^{(d)}_{\Delta},\ B^{(1)}_{2\Delta}, \ldots, B^{(d)}_{2\Delta},\ \ldots,\ B^{(1)}_{t\Delta}, \ldots, B^{(d)}_{t\Delta}\right)^\top \in \mathbb{R}^{td}. \tag{13}$$

The optimal h-step-ahead forecast of component $j \in \{1, \ldots, d\}$, given $\mathcal{X}_{t,d}$, is (BYZ Eq. 15)

$$\hat{B}^{(j)}_{(t+h)\Delta|(1:t)} = \mathbb{E}\left[B^{(j)}_{(t+h)\Delta} \,\middle|\, \mathcal{X}_{t,d} \right] = \left(\gamma^j_{t,h}\right)^\top \Sigma_{t,d}^{-1}\, \mathcal{X}_{t,d}, \tag{14}$$

where $\Sigma_{t,d} \in \mathbb{R}^{td\times td}$ is the covariance matrix of $\mathcal{X}_{t,d}$, obtained by evaluating the kernel (8) on the sampling grid, and $\gamma^j_{t,h} \in \mathbb{R}^{td}$ is the vector of covariances between the target $B^{(j)}_{(t+h)\Delta}$ and every element of $\mathcal{X}_{t,d}$, also obtained from (8). In practice, the parameter vector $(\mathbf{H}, \boldsymbol{\sigma}, \boldsymbol{\rho})$ is replaced by its sample estimate.

For the bivariate case, BYZ establish three theoretical properties of the optimal forecast that motivate the empirical investigation of the multivariate model:

1. **Monotonicity in dimension.** Adding a second component to the information set cannot increase the mean-squared forecast error of the leading component. Multivariate modelling weakly dominates

univariate modelling under correct specification.

2. **No gain under equal Hurst exponents.** When $H_1 = H_2$, the mean-squared forecast error of the bivariate optimal forecast coincides with that of the univariate one. The cross-section contributes only when the components differ in roughness.

3. **No gain under zero cross-correlation.** When $\rho_{12} = 0$, the optimal bivariate forecast also reduces to the univariate one. Cross-component information enters the forecast only through the cross-correlation channel.

Together, these results identify the two ingredients that drive any forecasting gain from the multivariate specification: *heterogeneity* in the Hurst exponents and *non-trivial* cross-correlation between components. Both ingredients are empirically plausible for the MOEX cross-section, and we investigate how well this bivariate result extends to higher-dimensional bundles of assets.

3 Data

We build a realized-variance panel from intraday price data on MOEX equities. The sample runs from January 1, 2015 to July 7, 2025 — over ten calendar years that include the 2014–2015 sanctions episode, the COVID-19 shock of 2020, and the geopolitical shock of 2022.

3.1 MOEX Equity Panel

The panel comprises the 43 most liquid common equities listed on the Moscow Exchange for which a continuous one-minute price history exists over the full sample period. The choice of constituents is conditioned on availability : an asset is admitted if its one-minute bars history is complete from 2015-01-01 through 2025-07-07, in the sense that there are no prolonged gaps in bars history. Also filter that accounts for number of minute bars per day is used. These two filters implicitly filter out non-liquid stocks which positively influence purity of the dataset.

The raw one-minute OHLCV bars are aggregated to a 5-minute sampling frequency before any realized-variance computation. Five-minute returns are the default sampling frequency in the realized-volatility literature.

For each trading day t and each instrument we then construct the daily realized variance RV_t by summing squared 5-minute log-returns over the regular session, as in equation (3). The object of empirical analysis throughout is the log-realized variance $y_t = \log \mathrm{RV}_t$.

4 Empirical Framework

This section specifies the rolling forecast design, the loss criterion, the DM inference procedure, and the companion-selection rule used throughout Section 5.

4.1 Forecast Design

All forecasts are produced in a rolling-window manner. The estimation window is fixed at $W = 504$ trading days, which corresponds to approximately two calendar years. This matches the window length in BYZ and is long enough for stable parameter estimates.

At each forecasting date t, every model under consideration consumes the same information set $\mathcal{F}_t = \{y_{t-W+1}, \ldots, y_t\}$ of the past W log-realized-variance observations of the target asset. Direct forecasting is used rather than iterated forecasting: a separate model is estimated for each horizon h in the grid $\mathcal{H} = \{1, 2, \ldots, 30\}$.

4.2 Evaluation Criterion

The primary criterion used to rank competing forecasts throughout the paper is the QLIKE loss function. For a true log-realized-variance observation y and a candidate forecast $\hat{y}$, expressed in the same logarithmic scale, the per-observation QLIKE loss is

$$\ell\big(y, \hat{y}\big) = e^{\hat{y}-y} - (\hat{y} - y) - 1. \tag{15}$$

Equation (15) is non-negative, achieves its minimum value of zero at $\hat{y} = y$, and is convex in the forecast. In contrast to a squared-error loss in logs, the QLIKE loss is asymmetric: it penalizes under-forecasts of variance more heavily than over-forecasts, reflecting the asymmetric loss profile of risk-management applications in which under-stating variance is more costly than over-stating it.

The key property of the QLIKE loss for our purposes is that it belongs to the class of forecast-evaluation criteria identified by Patton (2011) as *robust to the use of an imperfect realized-variance proxy.* The realized-variance proxy is a noisy measurement of the latent integrated variance, and many loss functions induce rankings of competing volatility forecasts that depend on the noise distribution of the proxy and can therefore reverse between proxies. Only loss functions belonging to a small family identified by Patton (2011) guarantee that the ranking of forecasts in expectation under the true (latent) variance coincides with the ranking under the proxy. The QLIKE loss is the most widely used member of that family in the volatility literature.

By contrast, the mean-squared forecast error (MSFE) of the log-realized variance does not satisfy the robustness property of Patton (2011). The use of MSFE for the empirical comparison in BYZ therefore leaves open the possibility that the reported gains reverse, or merely scale differently, under a robust criterion. We verify the forecast comparisons under QLIKE and retain MSFE only as a cross-check.

As a secondary, scale-free measure of forecast quality we report the out-of-sample coefficient of determination,

$$R^2_{i,h} = 1 - \frac{\sum_t \big(y_{i,t+h} - \hat{y}^{(m)}_{i,t+h}\big)^2}{\sum_t \big(y_{i,t+h} - \bar{y}_{i,h}\big)^2}, \tag{16}$$

where $\bar{y}_{i,h}$ denotes the out-of-sample mean of $y_{i,t+h}$. The R^2 statistic measures the share of out-of-sample variation of log-realized variance that the model explains and is useful for gauging the absolute predictive content of a forecast, as distinct from its relative performance against a competing model. We retain R^2 as a diagnostic because it provides a transparent scale on which the practical informativeness of a forecast can be judged.

For pairwise comparison between two competing forecasts m and m' we report the percentage QLIKE gain of m' over m,

$$\text{gain}_{i,h}^{\text{QLIKE}}(m \to m') = -\frac{\bar{\ell}_{i,h}^{(m')} - \bar{\ell}_{i,h}^{(m)}}{\bar{\ell}_{i,h}^{(m)}} \times 100\%, \tag{17}$$

where $\bar{\ell}_{i,h}^{(m)}$ denotes the mean QLIKE loss of model m on asset i at horizon h over the out-of-sample period. The analogous percentage MSFE gain is

$$\text{gain}_{i,h}^{\text{MSFE}}(m \to m') = -\frac{\overline{\text{MSFE}}_{i,h}^{(m')} - \overline{\text{MSFE}}_{i,h}^{(m)}}{\overline{\text{MSFE}}_{i,h}^{(m)}} \times 100\%, \tag{18}$$

where $\overline{\text{MSFE}}_{i,h}^{(m)} = \frac{1}{n}\sum_t \big(y_{i,t+h} - \hat{y}_{i,t+h}^{(m)}\big)^2$ is the mean squared forecast error of model m.

4.3 Statistical Inference on Forecast Differentials

Average-loss comparisons of competing forecasts are point estimates of expected-loss differentials and convey no information about the sampling uncertainty of those differentials. In a rolling-window forecasting design, where the same observations enter the estimation windows of adjacent forecast dates, the per-observation loss differentials are serially correlated, and naive standard errors understate the true uncertainty. We therefore accompany every pairwise model comparison with the formal test of Diebold and Mariano (1995) (DM), implemented with a Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) long-run variance estimator.

Concretely, for any pair of forecasts (m, m') and for a fixed horizon h, define the per-observation QLIKE differential

$$d_t^{(m,m')} = \ell\big(y_{t+h}, \hat{y}_{t+h}^{(m)}\big) - \ell\big(y_{t+h}, \hat{y}_{t+h}^{(m')}\big). \tag{19}$$

Under the null of equal predictive accuracy, the population mean of $d_t^{(m,m')}$ is zero. The DM test statistic is the sample mean of $d_t^{(m,m')}$ divided by a HAC estimate of its long-run standard error,

$$\text{DM} = \frac{\bar{d}^{(m,m')}}{\sqrt{\hat{V}_{\text{HAC}}\big(d^{(m,m')}\big)/n}}, \tag{20}$$

where n is the number of forecast dates in the out-of-sample period and $\hat{V}_{\text{HAC}}$ is a Newey–West estimator with bandwidth chosen as $\lfloor 1.5\,h \rfloor$. Under the null, the test statistic is asymptotically standard normal; given the out-of-sample size of $n \approx 2100$ observations, we refer it to the standard normal distribution.

We classify each pairwise comparison as follows: a *win* for model m' over m if the test rejects H_0 at $\alpha = 0.05$ and $\bar{d}^{(m,m')} > 0$ (model m' has lower mean loss); a *win* for m if the test rejects and $\bar{d}^{(m,m')} < 0$; and a *tie* otherwise. The *not worse* fraction for a model is the combined share of wins and ties for that model—equivalently, one minus the win share of the competing model.

4.4 Asset Selection for the Multivariate Forecast

The optimal forecast under a multivariate fBm model depends, beyond the target asset, on the choice of companion assets that enter the joint information set. BYZ select companion assets arbitrarily, a choice that bears no relation to the theoretical condition the paper itself derives. Our main methodological contribution is to close this gap by selecting companions according to a criterion that uses BYZ's own optimality condition.

For a target asset i, a candidate companion j, a forecast date t, and a forecast horizon h, define the in-sample Hurst gap $\Delta\hat{H}_{i,j,t} = |\hat{H}_{i,t} - \hat{H}_{j,t}|$ and the in-sample cross-correlation $\hat{\rho}_{i,j,t}$, both estimated from the data of the current rolling window $\mathcal{F}_t$ using the moment estimators of Section 2.4.4. We then rank candidates $j \in \mathcal{U} \setminus \{i\}$ by a scalar selection score $s_{i,j,t}$ and select the top $d-1$ candidates as the companion set, where d is a hyperparameter fixed in advance for each empirical exercise.

The selection score is the product:

$$s_{i,j,t} = \left|\hat{H}_{i,t} - \hat{H}_{j,t}\right| \cdot \left|\hat{\rho}_{i,j,t}\right|, \tag{21}$$

which captures, in its simplest non-trivial functional form, the qualitative shape of the BYZ optimality condition: gains from the multivariate forecast arise from the joint presence of heterogeneity in the Hurst exponents and non-trivial cross-correlation, so a useful score must be non-negligible on both ingredients simultaneously. The product form ensures that a candidate j scores well only when both $|\Delta\hat{H}_{i,j,t}|$ and $|\hat{\rho}_{i,j,t}|$ are non-negligible; a candidate with a sizable Hurst gap but a near-zero cross-correlation is ranked low, and conversely a candidate with a high cross-correlation but a Hurst exponent close to that of the target is also ranked low.

5 Empirical Results

5.1 Experiment 1: Univariate Fractional Brownian Motion versus HAR

Experiment 1 follows the methodology of Section 4.1 and evaluates whether modelling log-realized variance as a univariate fractional Brownian motion produces out-of-sample forecast improvements over the HAR benchmark of Corsi (2009). The exercise is conducted asset-by-asset on the MOEX cross-section. The main question here is: under the QLIKE loss, is the fBm framework worth deploying at all, and for which assets?

5.1.1 Average Performance Across Horizons

Table 1 reports the cross-sectional mean QLIKE and MSFE, the corresponding percentage gains of fBm over HAR, the share of tickers at which the per-ticker DM test rejects equality in favour of each model, and the out-of-sample R^2 differential, for a selection of horizons. The full table covering every horizon $h = 1, \dots, 30$ is reported in Table 5 in Appendix A.

h	QLIKE HAR	QLIKE fBm	QLIKE gain(%)	not worse (α=0.05) fBm	not worse (α=0.05) HAR	MSFE HAR	MSFE fBm	MSFE gain(%)	R^2 HAR	R^2 fBm
1	0.223	0.228	−2.2	0.63	0.93	0.438	0.441	−0.7	0.513	0.509
5	0.331	0.339	−2.5	0.79	0.95	0.642	0.644	−0.3	0.283	0.280
10	0.394	0.400	−1.5	0.91	0.98	0.744	0.741	0.4	0.168	0.170
15	0.433	0.432	0.3	0.93	0.95	0.797	0.785	1.5	0.110	0.122
22	0.471	0.467	0.9	0.98	0.95	0.844	0.816	3.3	0.059	0.088
30	0.507	0.502	1.1	1.00	0.93	0.894	0.853	4.5	0.004	0.045

Table 1: Cross-sectional summary of Experiment 1 at selected horizons. *gain* (%) is the percentage QLIKE (or MSFE) gain of fBm relative to HAR. The *not worse ($\alpha = 0.05$)* columns report, for each model, the combined share of tickers where that model wins or ties under the DM test (see Section 4.3): the fBm column is the fraction where either fBm has significantly lower loss than HAR, or neither difference is significant; the HAR column is defined symmetrically. The right-most two columns are the cross-sectional mean out-of-sample R^2 of HAR and fBm.

Table suggest that with respect to QLIKE loss HAR model dominates the univariate fBm at short horizons. At $h = 1$ HAR's average loss is roughly two percentage points lower. The HAR advantage shrinks monotonically with the horizon and reverses near $h = 15$, where the QLIKE gain turns positive. From there on the fBm advantage grows, but does so at horizons at which R^2 is already low for both models, as discussed below. Second, with respect to the out-of-sample R^2, the qualitative pattern is similar but the crossover occurs earlier. The fBm's R^2 appears to exceed that of HAR from approximately $h = 6$ onward.

One caveat: the R^2 of both models deteriorates rapidly with the horizon. At $h = 1$ the cross-sectional mean HAR R^2 is approximately 0.51; by $h = 15$ it falls to 0.11, and by $h = 22$ to roughly 0.06. Below an R^2 of approximately 0.12, which corresponds to $h \gtrsim 15$ on the present sample, the absolute predictability is low for both models. While the *relative* ranking favours fBm at these horizons, the practical significance of the comparison is limited. The most relevant competitive range is $h \in \{1, \dots, 14\}$, the region in which both models retain meaningful predictive content. Within this range HAR holds the edge on average, and the question becomes whether that edge is uniform across the cross-section or driven by a particular subset of assets.

5.1.2 Cross-Sectional Heterogeneity

Figure 2 shows the per-ticker, per-horizon percentage QLIKE gain of fBm relative to HAR as a heatmap.

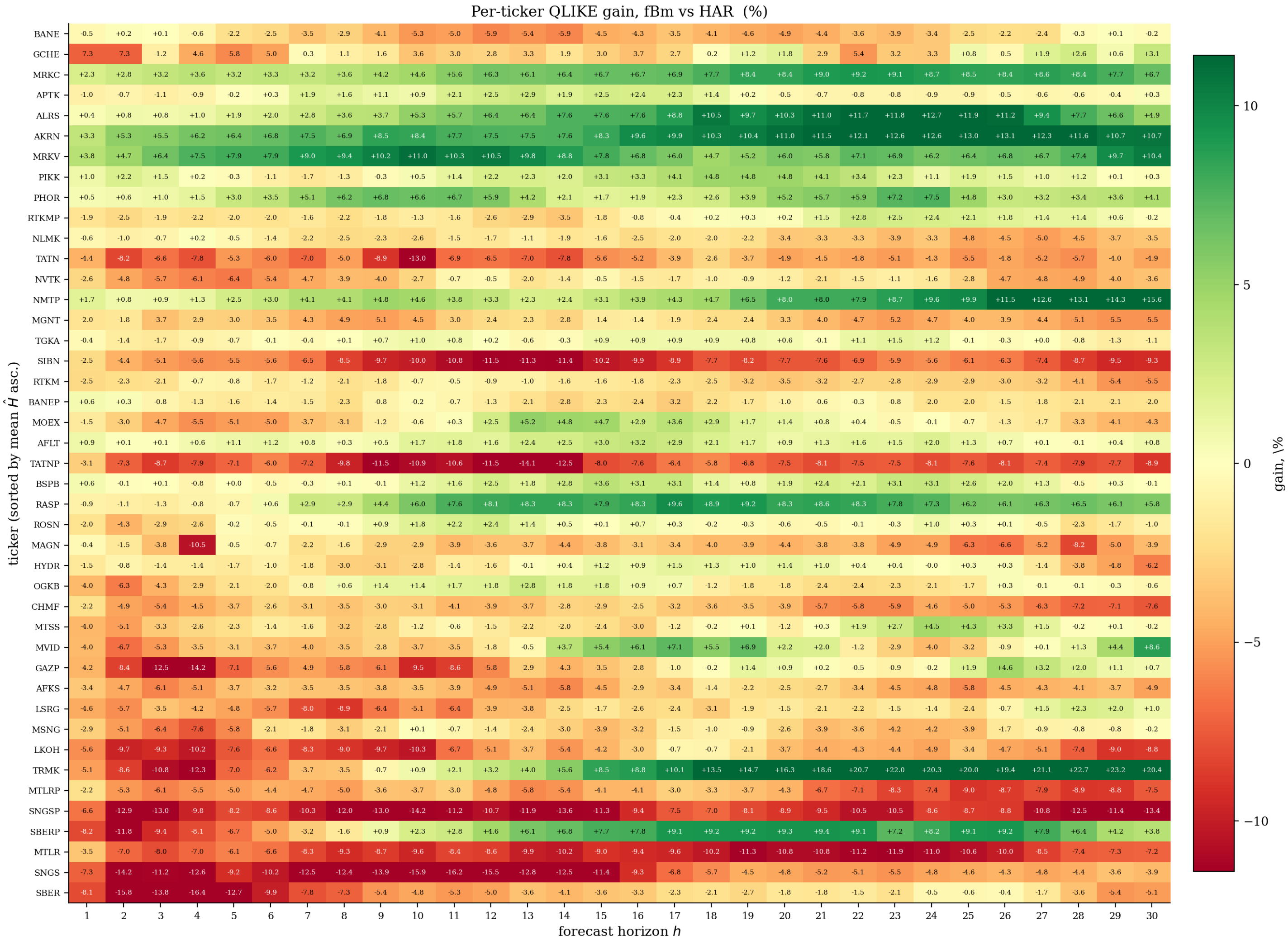


Figure 2: Heatmap of the percentage QLIKE gain of fBm relative to HAR for each of the 43 MOEX equities (rows) and for horizons $h = 1, \ldots, 30$ (columns). Green indicates that fBm outperforms HAR; red indicates the opposite.

First, the pattern of Section 5.1.1 holds at the per-asset level: red dominates at short horizons and green at long horizons for most assets. Second, there is visible clustering: one group of assets is predominantly green even at short horizons and another is predominantly red. Given that rows of the heatmap are sorted in ascending order by mean Hurst exponent(higher position correspond to lower H) it make sense to explore dependence of fBM gain over HAR under QLIKE loss in Hurst exponent of an asset.

5.1.3 The Role of the Hurst Exponent

To investigate the source of the cross-sectional heterogeneity, we regress, for each horizon h, the per-asset percentage QLIKE gain on the asset's time-series mean Hurst estimate $\bar{H}_i$ over the out-of-sample window:

$$\text{gain}_{i,h}^{\text{QLIKE}} = \alpha_h + \beta_h \, \bar{H}_i + u_{i,h}, \tag{22}$$

estimated by ordinary least squares with HC3-robust standard errors. Figure 3 displays the scatter plot and the fitted regression line at $h = 1$.

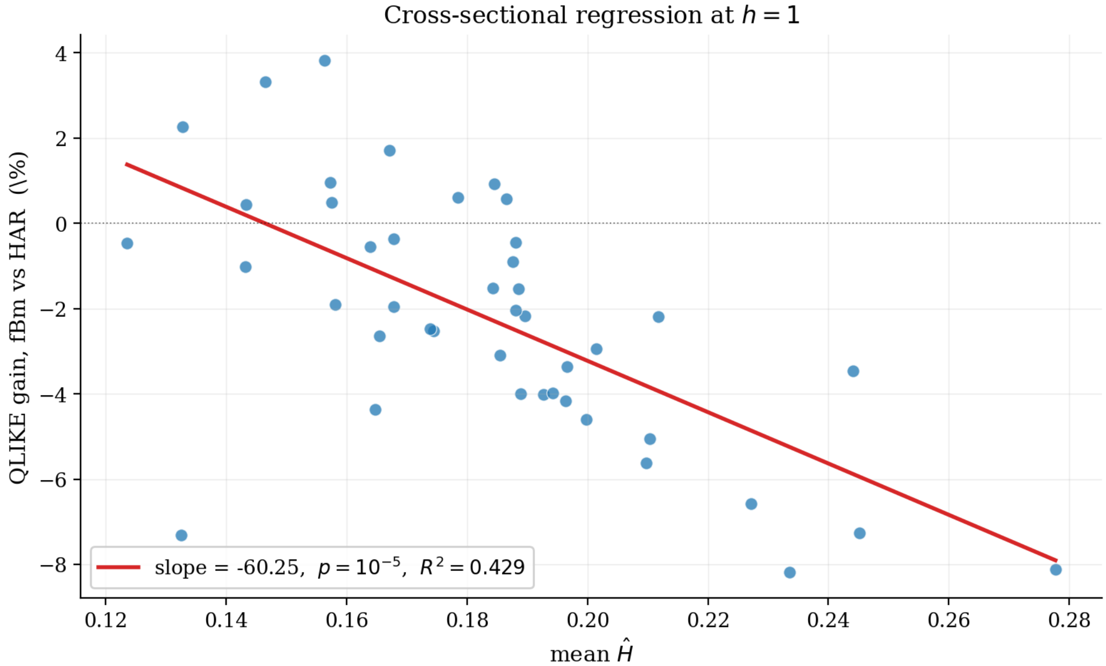
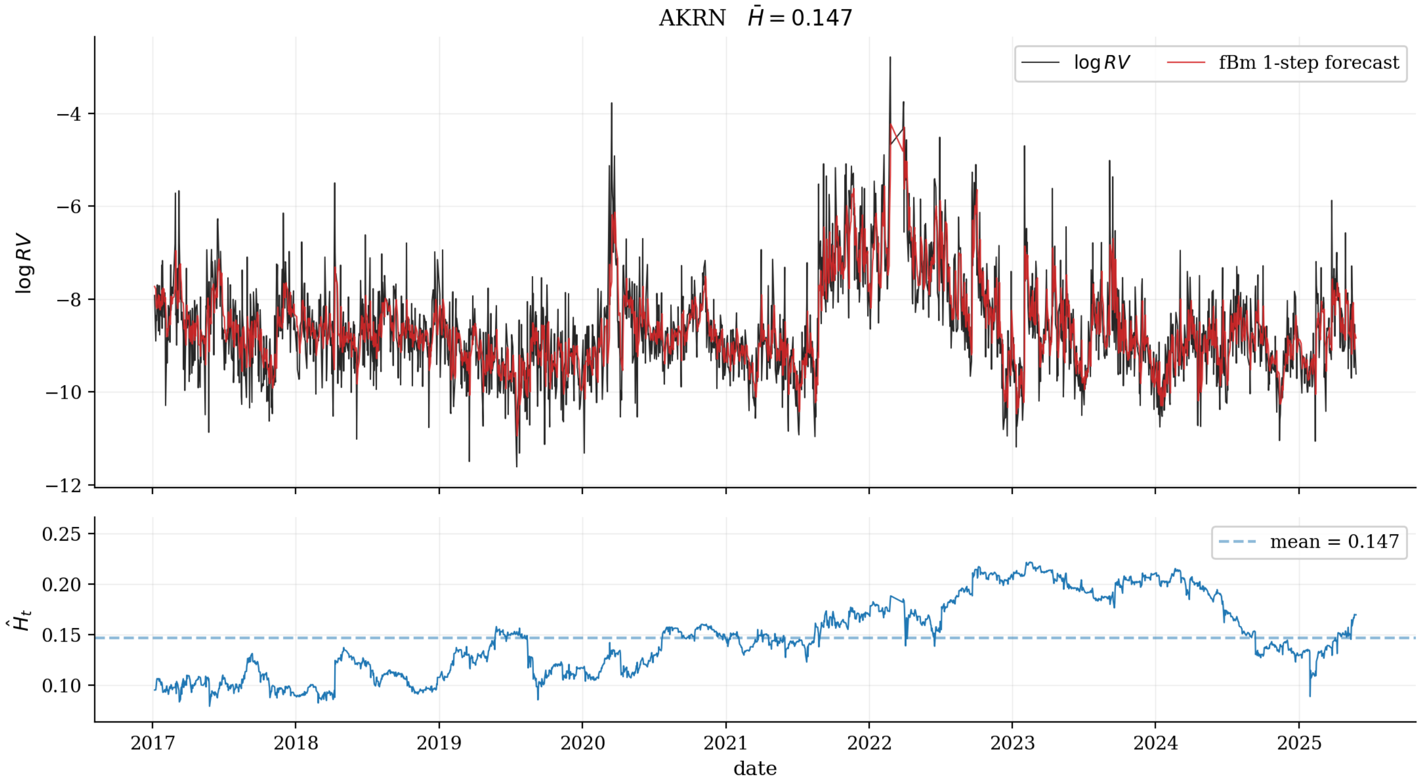


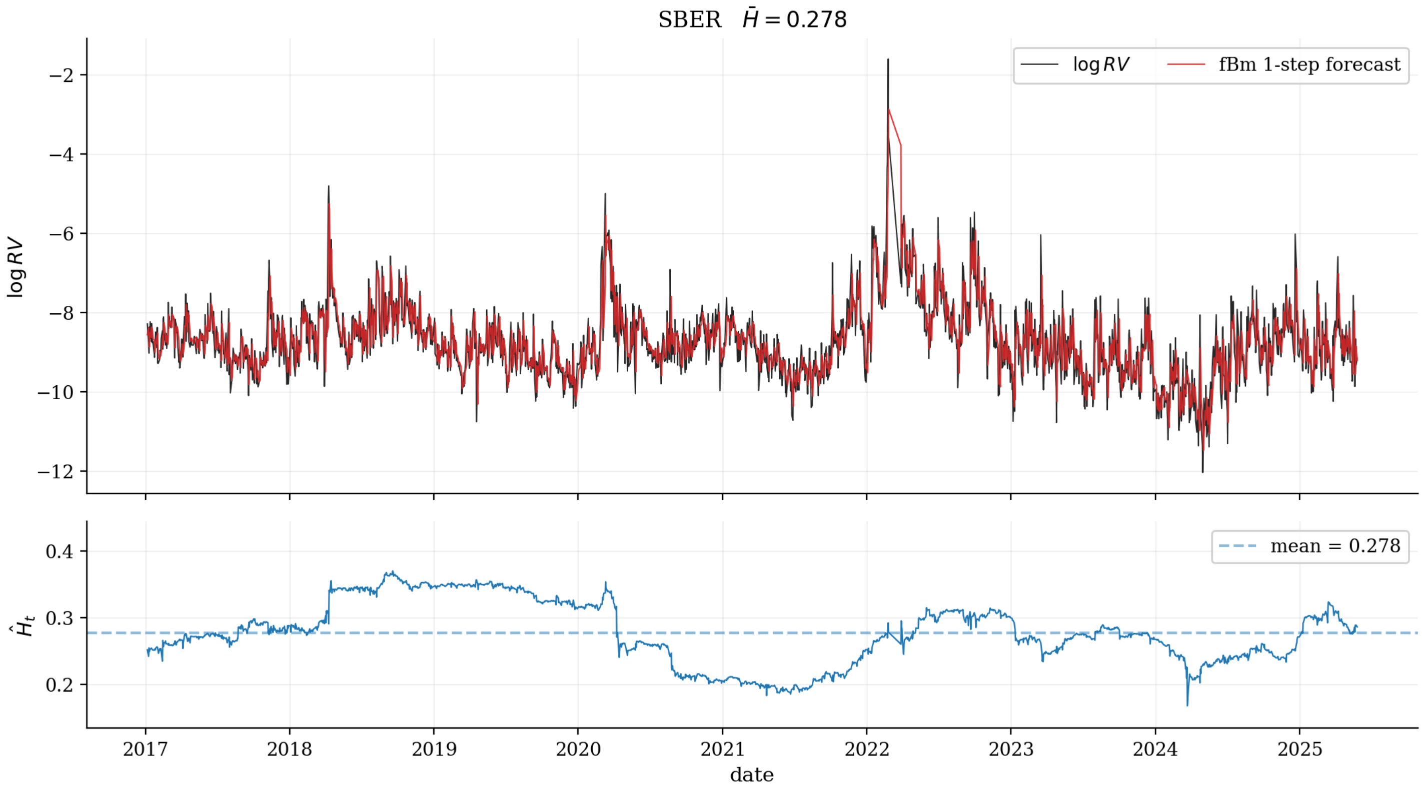
Figure 3: Cross-sectional OLS regression of the percentage QLIKE gain of fBm over HAR on the mean Hurst estimate $\bar{H}$ at horizon $h = 1$. Each dot is one of the 43 MOEX equities; standard errors are HC3-robust.

The slope is strongly negative: at $h = 1$, $\hat{\beta}_1 \approx -60.3$ ($p < 0.001$, $R^2 \approx 0.43$). Assets with lower mean Hurst exponent—that is, those whose realized volatility process is locally most irregular—benefit most from the fBm forecast. For assets with $\bar{H} \lesssim 0.16$ the fBm achieves a positive QLIKE gain even at the one-day horizon. For assets with $\bar{H} \gtrsim 0.22$ the HAR model is superior across virtually all short horizons.

To illustrate the contrast at the asset level, Figure 4 plots, for two representative assets, the rolling Hurst estimate $\hat{H}_t$ alongside the realized log-variance and the one-step-ahead fBm forecast. The two assets are AKRN ($\bar{H} \approx 0.147$), a low-$\bar{H}$ stock at which fBm is particularly competitive, and SBER ($\bar{H} \approx 0.278$), a high-$\bar{H}$ stock at which HAR tends to dominate.



(a) AKRN: low mean Hurst exponent ($\bar{H} \approx 0.147$).



(b) SBER: high mean Hurst exponent ($\bar{H} \approx 0.278$).

Figure 4: Rolling Hurst estimate $\hat{H}_t$ and log-realized variance with the fBm one-step-ahead forecast for two representative assets.

Table 2 reports the per-horizon QLIKE and MSFE comparison for both representative assets, together with the DM statistic, the p-value, and the verdict label (winner if the test rejects equality at $\alpha = 0.05$; – otherwise). The contrast is sharp: for AKRN the DM test rejects in favour of fBm at the majority of horizons under QLIKE, while for SBER it rejects in favour of HAR at every short horizon and never rejects in favour of fBm.

	QLIKE					MSFE				
h	HAR	fBm	DM	p	winner	HAR	fBm	DM	p	winner
1	0.3239	0.3131	3.50	0.0005	fBm	0.6132	0.6074	1.18	0.2363	–
5	0.4826	0.4516	3.11	0.0019	fBm	0.8605	0.8339	2.05	0.0407	fBm
10	0.5713	0.5234	2.23	0.0255	fBm	0.9751	0.9329	1.94	0.0526	–
15	0.6258	0.5740	1.84	0.0663	–	1.0480	0.9969	1.91	0.0561	–
22	0.6808	0.5983	2.95	0.0032	fBm	1.1206	1.0478	2.20	0.0277	fBm
30	0.7345	0.6557	2.50	0.0126	fBm	1.2112	1.1069	1.75	0.0811	–

(a) AKRN, $\bar{H} \approx 0.147$ (high-roughness asset).

	QLIKE					MSFE				
h	HAR	fBm	DM	p	winner	HAR	fBm	DM	p	winner
1	0.1520	0.1643	-2.29	0.0223	HAR	0.3034	0.3117	-2.04	0.0416	HAR
5	0.2434	0.2744	-3.52	0.0004	HAR	0.5094	0.5294	-1.65	0.0987	–
10	0.2998	0.3142	-1.04	0.2977	–	0.6195	0.6269	-0.44	0.6589	–
15	0.3535	0.3661	-0.43	0.6675	–	0.6918	0.6855	0.22	0.8278	–
22	0.3998	0.4057	-0.18	0.8608	–	0.7496	0.7377	0.33	0.7395	–
30	0.4370	0.4594	-0.53	0.5950	–	0.8176	0.8034	0.34	0.7355	–

(b) SBER, $\bar{H} \approx 0.278$ (low-roughness asset).

Table 2: Per-horizon Diebold–Mariano tests of HAR against the univariate fBm forecast, for the low- and high-$\bar{H}$ anchor assets and a selection of horizons. Statistic and p-value follow the construction of Section 4.3. Winner column reports *HAR* or *fBm* when the test rejects at $\alpha = 0.05$ and the dash – otherwise. Full horizon grids are reported in Tables 6–7 in the appendix.

Figure 3 gives a clear answer to the question raised in Section 1.4: the fractional framework is worth deploying *conditional* on the asset being sufficiently rough, and it loses to HAR on assets whose log-volatility is closer to a long-memory regime.

5.2 Experiment 2: Multivariate fBm vs. Univariate fBm

Experiment 2 tests whether the multivariate generalization actually delivers the forecasting improvement derived theoretically by BYZ. For every MOEX equity as the anchor and every bundle size $d \in \{1, \ldots, 9\}$, we form the dynamic mfBm forecast — with companions chosen at each step by the product score $|\hat{H}_i - \hat{H}_j| \cdot |\hat{\rho}_{ij}|$ of Section 4.4 — and compare it directly against the univariate fBm forecast for the same anchor.

We report two views of the comparison. Section 5.2.1 summarises the cross-sectional point-estimate evidence: the share of anchors at which the sample MSFE of mfBm exceeds that of fBm, and the cross-sectional mean percentage MSFE gain. Section 5.2.2 turns to inference, classifying each (anchor, h, d) cell by a per-anchor Diebold–Mariano test. The two views lead to qualitatively different conclusions. We focus on MSFE throughout because the BYZ theoretical result is derived for that criterion.

5.2.1 Descriptive Comparison

Table 3 reports the two descriptive summaries. Panel A is the fraction of the 43 MOEX anchors at which the sample MSFE of the dynamic-bundle mfBm exceeds that of the univariate fBm (values below 0.50 mean mfBm wins more often in the cross-section). Panel B is the cross-sectional mean of the per-anchor percentage MSFE gain of mfBm over fBm; positive values indicate that mfBm is on average better.

$h \backslash d$	1	2	3	4	5	6	7	8	9
Panel A: fraction of anchors with $MSFE_{\mathrm{mfBm}_d} > MSFE_{\mathrm{fBm}}$									
1	0.40	0.44	0.51	0.63	0.63	0.67	0.67	0.70	0.72
5	0.30	0.30	0.37	0.35	0.37	0.44	0.47	0.56	0.58
10	0.19	0.19	0.28	0.26	0.30	0.26	0.23	0.21	0.23
15	0.35	0.28	0.30	0.26	0.23	0.23	0.23	0.21	0.21
22	0.30	0.26	0.30	0.28	0.37	0.35	0.30	0.28	0.28
30	0.37	0.30	0.33	0.33	0.30	0.26	0.28	0.26	0.26
Panel B: mean MSFE gain mfBm vs fBm (%); negative = mfBm worse on average									
1	0.04	-0.01	-0.10	-0.21	-0.28	-0.36	-0.40	-0.43	-0.48
5	0.10	0.14	0.11	0.11	0.08	0.04	0.01	-0.01	-0.04
10	0.18	0.27	0.30	0.33	0.35	0.34	0.34	0.37	0.36
15	0.14	0.23	0.28	0.36	0.37	0.39	0.42	0.45	0.46
22	0.09	0.19	0.25	0.28	0.27	0.26	0.26	0.28	0.29
30	0.13	0.31	0.40	0.45	0.46	0.47	0.50	0.55	0.59

Table 3: Sample-MSFE comparison of the dynamic-bundle mfBm against the univariate fBm by (d, h). Panel A: fraction of the 43 MOEX anchors at which $\mathrm{MSFE}_{\mathrm{mfBm}_d} > \mathrm{MSFE}_{\mathrm{fBm}}$. Panel B: cross-sectional mean percentage MSFE gain of mfBm over fBm; positive values indicate that mfBm is on average better.

From the table it could be seen:

Horizon-conditional sign. At $h = 1$ the cross-sectional mean MSFE gain is negative for every $d \geq 2$ and grows worse with the bundle size, from -0.01% at $d = 2$ to -0.48% at $d = 9$. At every $h \geq 5$ the mean gain is positive for every d, and at horizons within the competitive range it rises with d, reaching $+0.42\%$ at $(h, d) = (14, 9)$.

Cross-sectional majority favours mfBm at $h > 5$. At every $h > 5$ the deterioration rate of Panel A falls below 0.50 for every bundle size: more anchors are better under mfBm than under fBm. At $h = 10$ the deterioration rate sits in the $[0.19, 0.30]$ range across d, meaning that mfBm has a lower sample MSFE for roughly seventy to eighty percent of MOEX anchors.

Bundle-size effects are not monotonic. At $h = 1$ adding partners is strictly harmful: both the deterioration rate and the mean gain move against mfBm as d grows. At $h \approx 10$ the dependence on d is essentially flat. At $h \geq 15$ the mean gain rises monotonically in d, in line with the BYZ statement that adding informative partners should help. The empirical analogue of the BYZ monotonicity in dimension therefore holds only at long horizons.

The point estimates of Table 3 are descriptive only. They show a systematic asymmetry in favour of mfBm at long horizons and against it at short horizons, but do not establish whether the asymmetry is statistically significant on a per-anchor basis.

5.2.2 Per-Anchor Diebold–Mariano Tests

For every (anchor, h, d) cell we run a Diebold–Mariano test of fBm against mfBm$_d$, treating the univariate fBm as the base. Two complementary fractions per (d, h) cell summarise the test: the share of anchors at which DM rejects in favour of mfBm (a significant improvement) and the share at which DM rejects in favour of fBm (a significant deterioration). Table 4 reports both fractions under QLIKE (Panels A and B) and the deterioration fraction under MSFE (Panel C).

$h\backslash d$	1	2	3	4	5	6	7	8	9
Panel A: fraction of anchors where DM rejects in favour of mfBm (QLIKE)									
1	0.05	0.09	0.02	0.02	0.02	0.02	0.02	0.02	0.02
5	0.09	0.12	0.09	0.07	0.07	0.07	0.05	0.02	0.00
10	0.02	0.07	0.02	0.02	0.02	0.02	0.02	0.02	0.00
15	0.00	0.07	0.02	0.00	0.00	0.00	0.00	0.00	0.00
22	0.00	0.05	0.02	0.00	0.00	0.00	0.00	0.00	0.00
30	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Panel B: fraction of anchors where DM rejects in favour of fBm (QLIKE) – significant deterioration									
1	0.00	0.00	0.00	0.09	0.07	0.09	0.12	0.12	0.12
5	0.05	0.00	0.00	0.02	0.02	0.02	0.00	0.00	0.00
10	0.02	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00
15	0.02	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.02
22	0.00	0.00	0.00	0.02	0.00	0.02	0.02	0.00	0.02
30	0.07	0.02	0.00	0.00	0.02	0.07	0.05	0.02	0.02
Panel C: fraction of anchors where DM rejects in favour of fBm (MSFE) – significant deterioration									
1	0.05	0.05	0.07	0.14	0.19	0.21	0.23	0.30	0.30
5	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
30	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 4: Per-anchor Diebold–Mariano tests of fBm against the dynamic-bundle mfBm at $\alpha = 0.05$ (selected horizons). Panel A: share of the 43 MOEX anchors at which DM rejects in favour of mfBm under QLIKE. Panel B: share at which DM rejects in favour of fBm under QLIKE (significant deterioration). Panel C: share at which DM rejects in favour of fBm under MSFE.

The inferential picture is much weaker than the descriptive one.

Most anchors are statistically indistinguishable. Across the (d, h) grid the share of anchors at which the per-anchor DM test rejects in either direction never exceeds twelve percent under QLIKE. For the overwhelming majority of MOEX anchors — between 86% and 100% in every cell — the data cannot reject equality of the two forecasts at the 5% level.

Significant rejections favour mfBm at $h \geq 5$. At $h = 5$–15, Panel A consistently exceeds Panel B: significant rejections, when they do occur, are more often in favour of mfBm. The asymmetry is modest — the gap between the two panels never exceeds ten percentage points — but it appears to be consistent in direction across the relevant horizons. Combined with the descriptive evidence, the picture at $h \geq 5$ is:

mfBm rarely wins in a statistically significant sense, but when it does, the direction is consistent.

Summary

The data of Experiment 2 support two conclusions.

1. **Point estimates favour mfBm at $h > 5$.** The cross-sectional mean MSFE gain is positive for every bundle size and rises with both bundle size and horizon, reaching +0.42% at $(h, d) = (14, 9)$; at $h = 10$ roughly seventy to eighty percent of MOEX anchors have a lower sample MSFE under mfBm.

2. **Per-anchor DM rarely reaches significance.** For 86–100% of anchors at any (d, h), the test cannot reject equality of the two forecasts. The case for mfBm is cross-sectional: it rests on the systematic small gain and on the asymmetric direction of the rare significant rejections, not on anchor-by-anchor dominance.

6 Conclusion

We revisit the multivariate rough-volatility framework of BYZ (Bibinger et al., 2025) on a new market and under three methodological refinements. The evidence from 43 MOEX equities over 2015–2025 supports each of the four contributions of Section 1.4, with varying strength.

Under the proxy-robust QLIKE loss, the qualitative finding of BYZ (Bibinger et al., 2025) survives: the fractional family overtakes the HAR benchmark of Corsi (2009) near $h = 15$, though, as Section 5.1.1 notes, absolute predictability is already modest in both models at that horizon, so the practical significance of the reversal is limited. Importantly, the choice of metric matters for the timing of the crossover: under MSFE the cross-sectional mean gain of fBm over HAR turns positive at $h \approx 6$, whereas under the proxy-robust QLIKE the crossover is delayed to $h \approx 15$. Reporting under MSFE alone, as in BYZ, would credit the fractional framework with an advantage at horizons at which it has none under the proxy-robust criterion.

Inference matters at the per-asset level. The Diebold–Mariano test with Bartlett HAC (Section 4.3) reveals that the cross-sectional average gain hides material heterogeneity: while the HAR-significantly-wins share collapses monotonically from 37% at $h = 1$ to 9% by $h = 10$ and stays around 7% through $h = 14$, the fBm-significantly-wins share remains in a narrow 2–7% band across horizons.

A cross-sectional OLS regression of the QLIKE gain on the time-series mean Hurst estimate $\bar{H}_i$ gives a slope of -60.3 ($p < 0.001$, $R^2 = 0.43$) at the one-day horizon. The contrast between AKRN ($\bar{H} \approx 0.147$) and SBER ($\bar{H} \approx 0.278$) illustrates this: the DM test rejects in favour of fBm at the majority of horizons for the former and in favour of HAR at every short horizon for the latter, with no rejection in favour of fBm on SBER at any horizon.

The theory-consistent dynamic companion-selection rule of Section 4.4 produces a *systematic but small* cross-sectional improvement over the univariate fBm at $h \geq 5$. The cross-sectional mean MSFE gain of mfBm$_d$ over the univariate fBm rises from +0.12% at $d = 1$ to +0.42% at $d = 9$ at $h = 14$, and at

$h = 10$ roughly seventy to eighty percent of MOEX anchors have a lower sample MSFE under mfBm than under fBm. The per-anchor improvement is, however, rarely large enough to register on Diebold–Mariano tests: for 86–100% of anchors at any (d, h), the two forecasts are statistically indistinguishable at $\alpha = 0.05$. The empirical case for the multivariate model is therefore cross-sectional rather than per-anchor: it rests on the systematic small advantage in expectation and on an asymmetric direction of the rare significant rejections, not on anchor-by-anchor dominance. The BYZ-theoretical monotonicity in dimension correspondingly holds approximately in expectation at long horizons but fails at short horizons.

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A Full Result Tables

The compact summaries reported in Section 5 cover a selection of horizons. The full tables for every horizon $h \in \{1, \dots, 30\}$ are listed below.

A.1 Cross-Sectional Summary, Experiment 1

	QLIKE			not worse (α=0.05)		MSFE			R^2	
h	HAR	fBm	gain(%)	fBm	HAR	HAR	fBm	gain(%)	HAR	fBm
1	0.223	0.228	−2.2	0.63	0.93	0.438	0.441	−0.7	0.513	0.509
2	0.272	0.282	−3.6	0.53	0.95	0.534	0.540	−1.2	0.405	0.398
3	0.297	0.307	−3.4	0.60	0.98	0.580	0.587	−1.1	0.352	0.344
4	0.316	0.327	−3.6	0.70	0.95	0.614	0.618	−0.7	0.315	0.309
5	0.331	0.339	−2.5	0.79	0.95	0.642	0.644	−0.3	0.283	0.280
6	0.346	0.353	−2.2	0.91	0.95	0.666	0.666	0.1	0.255	0.256
7	0.360	0.367	−1.9	0.86	0.98	0.690	0.688	0.2	0.228	0.230
8	0.372	0.380	−2.1	0.86	0.98	0.712	0.711	0.1	0.204	0.204
9	0.385	0.391	−1.7	0.93	0.98	0.730	0.729	0.2	0.183	0.184
10	0.394	0.400	−1.5	0.91	0.98	0.744	0.741	0.4	0.168	0.170
11	0.404	0.408	−1.0	0.93	0.98	0.758	0.753	0.6	0.152	0.157
12	0.411	0.414	−0.8	0.93	0.98	0.769	0.765	0.6	0.139	0.144
13	0.419	0.422	−0.7	0.93	0.98	0.780	0.774	0.7	0.128	0.133
14	0.426	0.428	−0.5	0.93	0.95	0.788	0.780	1.0	0.120	0.127
15	0.433	0.432	0.3	0.93	0.95	0.797	0.785	1.5	0.110	0.122
16	0.440	0.438	0.4	0.95	0.95	0.804	0.790	1.7	0.103	0.117
17	0.445	0.441	1.0	0.98	0.95	0.811	0.794	2.1	0.095	0.112
18	0.451	0.445	1.3	0.98	0.95	0.818	0.799	2.4	0.087	0.107
19	0.457	0.451	1.4	0.98	0.95	0.826	0.805	2.5	0.079	0.100
20	0.461	0.456	1.2	0.98	0.95	0.831	0.809	2.7	0.073	0.095
21	0.467	0.462	1.0	0.98	0.95	0.838	0.813	3.0	0.065	0.091
22	0.471	0.467	0.9	0.98	0.95	0.844	0.816	3.3	0.059	0.088
23	0.476	0.472	0.9	1.00	0.95	0.851	0.821	3.5	0.052	0.083
24	0.480	0.476	0.9	1.00	0.98	0.858	0.825	3.9	0.043	0.078
25	0.486	0.481	0.9	1.00	0.98	0.866	0.830	4.2	0.035	0.073
26	0.489	0.483	1.1	1.00	0.98	0.871	0.831	4.5	0.029	0.071
27	0.493	0.487	1.1	1.00	0.98	0.876	0.836	4.6	0.023	0.065
28	0.498	0.494	0.9	1.00	0.98	0.883	0.844	4.5	0.015	0.056
29	0.502	0.497	1.0	1.00	0.95	0.888	0.849	4.5	0.010	0.050
30	0.507	0.502	1.1	1.00	0.93	0.894	0.853	4.5	0.004	0.045

Table 5: Full $h = 1, \dots, 30$ cross-sectional summary of HAR vs. fBm under QLIKE and MSFE, with the four statistical-significance fractions per metric. Compact summary in Table 1.

A.2 Per-Anchor Tables, Experiment 1

	QLIKE					MSFE				
h	HAR	fBm	DM	p	winner	HAR	fBm	DM	p	winner
1	0.3239	0.3131	3.50	0.0005	fBm	0.6132	0.6074	1.18	0.2363	–
2	0.3841	0.3636	4.23	0.0000	fBm	0.7195	0.7065	1.64	0.1011	–
3	0.4187	0.3956	3.79	0.0002	fBm	0.7732	0.7535	2.00	0.0461	fBm
4	0.4514	0.4234	3.60	0.0003	fBm	0.8184	0.7923	2.22	0.0268	fBm
5	0.4826	0.4516	3.11	0.0019	fBm	0.8605	0.8339	2.05	0.0407	fBm
6	0.5075	0.4729	3.02	0.0025	fBm	0.8862	0.8611	1.81	0.0700	–
7	0.5224	0.4833	2.81	0.0051	fBm	0.9159	0.8876	1.74	0.0820	–
8	0.5441	0.5067	2.26	0.0241	fBm	0.9393	0.9090	1.66	0.0969	–
9	0.5574	0.5100	2.54	0.0110	fBm	0.9544	0.9138	2.04	0.0418	fBm
10	0.5713	0.5234	2.23	0.0255	fBm	0.9751	0.9329	1.94	0.0526	–
11	0.5816	0.5368	1.77	0.0770	–	0.9905	0.9501	1.70	0.0889	–
12	0.5929	0.5483	1.69	0.0903	–	1.0110	0.9674	1.70	0.0886	–
13	0.6067	0.5614	1.60	0.1103	–	1.0258	0.9796	1.74	0.0818	–
14	0.6124	0.5660	1.63	0.1026	–	1.0321	0.9842	1.85	0.0643	–
15	0.6258	0.5740	1.84	0.0663	–	1.0480	0.9969	1.91	0.0561	–
16	0.6312	0.5703	2.36	0.0185	fBm	1.0559	0.9949	2.33	0.0200	fBm
17	0.6412	0.5777	2.60	0.0094	fBm	1.0629	1.0026	2.34	0.0192	fBm
18	0.6513	0.5841	2.77	0.0056	fBm	1.0769	1.0183	2.28	0.0229	fBm
19	0.6622	0.5936	2.75	0.0059	fBm	1.0949	1.0347	2.13	0.0337	fBm
20	0.6727	0.5987	2.86	0.0042	fBm	1.1062	1.0419	2.19	0.0286	fBm
21	0.6734	0.5963	2.99	0.0028	fBm	1.1124	1.0438	2.24	0.0249	fBm
22	0.6808	0.5983	2.95	0.0032	fBm	1.1206	1.0478	2.20	0.0277	fBm
23	0.6893	0.6024	2.95	0.0032	fBm	1.1310	1.0488	2.29	0.0224	fBm
24	0.6981	0.6104	2.86	0.0043	fBm	1.1439	1.0594	2.15	0.0317	fBm
25	0.7087	0.6168	2.85	0.0044	fBm	1.1594	1.0704	2.06	0.0396	fBm
26	0.7170	0.6231	2.93	0.0034	fBm	1.1677	1.0750	2.02	0.0431	fBm
27	0.7202	0.6315	2.94	0.0033	fBm	1.1777	1.0807	1.99	0.0469	fBm
28	0.7230	0.6388	2.74	0.0063	fBm	1.1877	1.0895	1.93	0.0537	–
29	0.7261	0.6487	2.48	0.0131	fBm	1.1979	1.0994	1.79	0.0743	–
30	0.7345	0.6557	2.50	0.0126	fBm	1.2112	1.1069	1.75	0.0811	–

Table 6: Full per-horizon DM table for AKRN ($\bar{H} \approx 0.147$): HAR vs. univariate fBm. Compact view in Table 2.

	QLIKE					MSFE				
h	HAR	fBm	DM	p	winner	HAR	fBm	DM	p	winner
1	0.1520	0.1643	−2.29	0.0223	HAR	0.3034	0.3117	−2.04	0.0416	HAR
2	0.2008	0.2325	−2.16	0.0307	HAR	0.4070	0.4258	−2.29	0.0221	HAR
3	0.2188	0.2490	−3.11	0.0019	HAR	0.4542	0.4777	−2.51	0.0121	HAR
4	0.2320	0.2702	−2.48	0.0134	HAR	0.4831	0.5064	−2.16	0.0310	HAR
5	0.2434	0.2744	−3.52	0.0004	HAR	0.5094	0.5294	−1.65	0.0987	–
6	0.2500	0.2747	−3.24	0.0012	HAR	0.5299	0.5470	−1.37	0.1711	–
7	0.2619	0.2823	−2.55	0.0109	HAR	0.5547	0.5704	−1.19	0.2352	–
8	0.2755	0.2955	−2.15	0.0316	HAR	0.5792	0.5939	−1.03	0.3012	–
9	0.2885	0.3041	−1.42	0.1552	–	0.6023	0.6126	−0.67	0.5043	–
10	0.2998	0.3142	−1.04	0.2977	–	0.6195	0.6269	−0.44	0.6589	–
11	0.3125	0.3291	−0.96	0.3382	–	0.6400	0.6481	−0.42	0.6752	–
12	0.3249	0.3411	−0.80	0.4258	–	0.6554	0.6613	−0.27	0.7889	–
13	0.3333	0.3454	−0.53	0.5991	–	0.6676	0.6703	−0.11	0.9145	–
14	0.3431	0.3571	−0.51	0.6104	–	0.6802	0.6780	0.08	0.9377	–
15	0.3535	0.3661	−0.43	0.6675	–	0.6918	0.6855	0.22	0.8278	–
16	0.3601	0.3719	−0.38	0.7028	–	0.7026	0.6953	0.24	0.8140	–
17	0.3674	0.3760	−0.28	0.7820	–	0.7144	0.7078	0.20	0.8399	–
18	0.3724	0.3801	−0.25	0.8039	–	0.7218	0.7119	0.30	0.7668	–
19	0.3801	0.3902	−0.32	0.7455	–	0.7302	0.7216	0.25	0.7988	–
20	0.3869	0.3937	−0.22	0.8285	–	0.7376	0.7260	0.34	0.7346	–
21	0.3932	0.4003	−0.21	0.8314	–	0.7435	0.7311	0.35	0.7238	–
22	0.3998	0.4057	−0.18	0.8608	–	0.7496	0.7377	0.33	0.7395	–
23	0.4074	0.4161	−0.24	0.8080	–	0.7566	0.7420	0.39	0.6952	–
24	0.4147	0.4170	−0.06	0.9486	–	0.7658	0.7435	0.58	0.5640	–
25	0.4256	0.4281	−0.07	0.9479	–	0.7778	0.7490	0.69	0.4932	–
26	0.4326	0.4343	−0.04	0.9666	–	0.7870	0.7541	0.75	0.4540	–
27	0.4359	0.4432	−0.17	0.8619	–	0.7958	0.7660	0.68	0.4942	–
28	0.4466	0.4627	−0.36	0.7226	–	0.8094	0.7880	0.49	0.6276	–
29	0.4369	0.4605	−0.54	0.5859	–	0.8133	0.7988	0.33	0.7406	–
30	0.4370	0.4594	−0.53	0.5950	–	0.8176	0.8034	0.34	0.7355	–

Table 7: Full per-horizon DM table for SBER ($\bar{H} \approx 0.278$): HAR vs. univariate fBm. Compact view in Table 2.

A.3 Full (d, h) Grids, Experiment 2

$h\backslash d$	1	2	3	4	5	6	7	8	9
1	0.40	0.44	0.51	0.63	0.63	0.67	0.67	0.70	0.72
2	0.40	0.40	0.49	0.49	0.56	0.65	0.72	0.74	0.77
3	0.33	0.28	0.40	0.44	0.44	0.53	0.53	0.60	0.70
4	0.30	0.30	0.30	0.37	0.42	0.53	0.56	0.63	0.65
5	0.30	0.30	0.37	0.35	0.37	0.44	0.47	0.56	0.58
6	0.26	0.28	0.30	0.26	0.26	0.28	0.35	0.37	0.44
7	0.26	0.30	0.33	0.35	0.28	0.35	0.40	0.40	0.42
8	0.26	0.19	0.23	0.16	0.23	0.28	0.23	0.23	0.23
9	0.21	0.23	0.26	0.21	0.26	0.26	0.23	0.21	0.21
10	0.19	0.19	0.28	0.26	0.30	0.26	0.23	0.21	0.23
11	0.28	0.21	0.28	0.26	0.28	0.23	0.23	0.21	0.21
12	0.19	0.16	0.19	0.16	0.21	0.23	0.21	0.23	0.21
13	0.33	0.23	0.28	0.23	0.30	0.30	0.33	0.26	0.26
14	0.33	0.30	0.30	0.33	0.33	0.28	0.30	0.28	0.23
15	0.35	0.28	0.30	0.26	0.23	0.23	0.23	0.21	0.21
16	0.35	0.26	0.33	0.26	0.28	0.26	0.26	0.23	0.21
17	0.33	0.26	0.26	0.23	0.26	0.26	0.28	0.26	0.23
18	0.35	0.21	0.23	0.23	0.26	0.26	0.26	0.23	0.26
19	0.33	0.28	0.28	0.23	0.30	0.26	0.30	0.26	0.26
20	0.33	0.28	0.40	0.33	0.37	0.33	0.30	0.28	0.30
21	0.30	0.28	0.35	0.37	0.37	0.35	0.35	0.33	0.37
22	0.30	0.26	0.30	0.28	0.37	0.35	0.30	0.28	0.28
23	0.33	0.28	0.33	0.35	0.33	0.30	0.30	0.30	0.23
24	0.28	0.23	0.33	0.33	0.33	0.28	0.28	0.23	0.26
25	0.28	0.33	0.35	0.35	0.35	0.35	0.33	0.33	0.30
26	0.30	0.28	0.33	0.30	0.30	0.30	0.26	0.23	0.23
27	0.30	0.30	0.35	0.26	0.28	0.28	0.23	0.19	0.19
28	0.35	0.33	0.35	0.35	0.28	0.28	0.26	0.21	0.19
29	0.35	0.33	0.33	0.33	0.33	0.30	0.28	0.26	0.23
30	0.37	0.30	0.33	0.33	0.30	0.26	0.28	0.26	0.26

Table 8: Sample-MSFE comparison of the dynamic-bundle mfBm against the univariate fBm, $h = 1, \ldots, 30$ and $d = 1, \ldots, 9$ — fraction of the 43 MOEX anchors at which $\mathrm{MSFE}_{\mathrm{mfBm}_d} > \mathrm{MSFE}_{\mathrm{fBm}}$. Compact view (Panel A of Table 3).

$h\backslash d$	1	2	3	4	5	6	7	8	9
1	0.04	−0.01	−0.10	−0.21	−0.28	−0.36	−0.40	−0.43	−0.48
2	0.05	0.05	−0.02	−0.09	−0.14	−0.20	−0.23	−0.25	−0.29
3	0.05	0.07	0.03	−0.02	−0.06	−0.12	−0.14	−0.17	−0.19
4	0.08	0.11	0.08	0.04	0.00	−0.06	−0.09	−0.12	−0.15
5	0.10	0.14	0.11	0.11	0.08	0.04	0.01	−0.01	−0.04
6	0.11	0.15	0.15	0.16	0.14	0.11	0.09	0.08	0.06
7	0.10	0.14	0.14	0.15	0.15	0.12	0.11	0.10	0.08
8	0.14	0.19	0.21	0.23	0.23	0.22	0.21	0.22	0.21
9	0.16	0.23	0.26	0.28	0.28	0.28	0.28	0.30	0.28
10	0.18	0.27	0.30	0.33	0.35	0.34	0.34	0.37	0.36
11	0.15	0.25	0.27	0.30	0.32	0.31	0.33	0.35	0.35
12	0.17	0.27	0.31	0.35	0.36	0.36	0.38	0.40	0.41
13	0.16	0.26	0.28	0.33	0.33	0.34	0.36	0.39	0.39
14	0.12	0.23	0.27	0.33	0.34	0.36	0.38	0.42	0.42
15	0.14	0.23	0.28	0.36	0.37	0.39	0.42	0.45	0.46
16	0.13	0.22	0.29	0.36	0.38	0.40	0.42	0.44	0.45
17	0.11	0.23	0.30	0.35	0.36	0.39	0.41	0.44	0.45
18	0.11	0.24	0.30	0.36	0.36	0.38	0.40	0.43	0.43
19	0.11	0.22	0.26	0.31	0.30	0.32	0.32	0.35	0.35
20	0.12	0.21	0.24	0.28	0.28	0.29	0.29	0.31	0.31
21	0.09	0.21	0.24	0.29	0.29	0.29	0.29	0.31	0.31
22	0.09	0.19	0.25	0.28	0.27	0.26	0.26	0.28	0.29
23	0.10	0.22	0.26	0.29	0.28	0.27	0.28	0.30	0.31
24	0.11	0.23	0.27	0.30	0.28	0.28	0.28	0.31	0.32
25	0.09	0.20	0.24	0.27	0.25	0.23	0.24	0.27	0.28
26	0.13	0.24	0.29	0.32	0.31	0.30	0.31	0.34	0.34
27	0.13	0.27	0.35	0.39	0.40	0.40	0.41	0.44	0.46
28	0.12	0.28	0.36	0.41	0.42	0.43	0.44	0.49	0.52
29	0.12	0.29	0.37	0.41	0.42	0.43	0.46	0.50	0.54
30	0.13	0.31	0.40	0.45	0.46	0.47	0.50	0.55	0.59

Table 9: Sample-MSFE comparison of the dynamic-bundle mfBm against the univariate fBm, $h = 1, \ldots, 30$ and $d = 1, \ldots, 9$ — cross-sectional mean percentage MSFE gain of mfBm over fBm; positive values indicate that mfBm is on average better. Compact view (Panel B of Table 3).

$h\backslash d$	1	2	3	4	5	6	7	8	9
1	0.05	0.09	0.02	0.02	0.02	0.02	0.02	0.02	0.02
2	0.05	0.07	0.05	0.00	0.05	0.02	0.05	0.02	0.02
3	0.05	0.07	0.05	0.02	0.02	0.02	0.02	0.02	0.02
4	0.05	0.09	0.05	0.02	0.05	0.02	0.00	0.00	0.00
5	0.09	0.12	0.09	0.07	0.07	0.07	0.05	0.02	0.00
6	0.16	0.16	0.14	0.09	0.09	0.05	0.05	0.05	0.02
7	0.12	0.14	0.09	0.09	0.05	0.02	0.02	0.02	0.00
8	0.12	0.14	0.09	0.05	0.02	0.02	0.02	0.02	0.02
9	0.05	0.07	0.07	0.05	0.05	0.05	0.02	0.02	0.00
10	0.02	0.07	0.02	0.02	0.02	0.02	0.02	0.02	0.00
11	0.02	0.05	0.02	0.02	0.02	0.02	0.02	0.02	0.00
12	0.09	0.07	0.07	0.05	0.02	0.02	0.02	0.02	0.02
13	0.07	0.09	0.07	0.00	0.02	0.00	0.00	0.00	0.00
14	0.00	0.07	0.05	0.00	0.00	0.00	0.00	0.00	0.00
15	0.00	0.07	0.02	0.00	0.00	0.00	0.00	0.00	0.00
16	0.02	0.09	0.02	0.00	0.00	0.00	0.00	0.00	0.00
17	0.00	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00
18	0.00	0.07	0.05	0.02	0.02	0.02	0.02	0.02	0.00
19	0.02	0.07	0.05	0.00	0.00	0.02	0.02	0.02	0.00
20	0.00	0.05	0.02	0.00	0.00	0.00	0.00	0.00	0.00
21	0.02	0.07	0.02	0.00	0.00	0.00	0.00	0.00	0.00
22	0.00	0.05	0.02	0.00	0.00	0.00	0.00	0.00	0.00
23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
26	0.02	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
30	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 10: Per-anchor Diebold–Mariano tests of fBm against the dynamic-bundle mfBm at $\alpha = 0.05$, $h = 1, \ldots, 30$ and $d = 1, \ldots, 9$ — share of the 43 MOEX anchors at which DM rejects in favour of mfBm under QLIKE. Compact view (Panel A of Table 4).

$h\backslash d$	1	2	3	4	5	6	7	8	9
1	0.00	0.00	0.00	0.09	0.07	0.09	0.12	0.12	0.12
2	0.05	0.00	0.00	0.05	0.05	0.05	0.07	0.09	0.09
3	0.05	0.00	0.00	0.00	0.02	0.05	0.05	0.07	0.07
4	0.00	0.02	0.00	0.00	0.02	0.02	0.00	0.02	0.02
5	0.05	0.00	0.00	0.02	0.02	0.02	0.00	0.00	0.00
6	0.02	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
7	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
10	0.02	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00
11	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
12	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14	0.02	0.02	0.00	0.00	0.02	0.02	0.02	0.02	0.02
15	0.02	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.02
16	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
17	0.00	0.02	0.00	0.02	0.02	0.00	0.00	0.00	0.00
18	0.00	0.00	0.02	0.02	0.02	0.00	0.00	0.00	0.00
19	0.00	0.00	0.02	0.02	0.02	0.00	0.00	0.00	0.00
20	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00
21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
22	0.00	0.00	0.00	0.02	0.00	0.02	0.02	0.00	0.02
23	0.00	0.00	0.00	0.02	0.00	0.02	0.02	0.02	0.02
24	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
25	0.02	0.00	0.00	0.02	0.02	0.02	0.02	0.02	0.02
26	0.02	0.00	0.00	0.02	0.00	0.02	0.05	0.00	0.00
27	0.07	0.00	0.00	0.02	0.05	0.05	0.02	0.02	0.02
28	0.05	0.00	0.05	0.02	0.05	0.05	0.05	0.02	0.02
29	0.07	0.00	0.00	0.02	0.02	0.02	0.02	0.02	0.02
30	0.07	0.02	0.00	0.00	0.02	0.07	0.05	0.02	0.02

Table 11: Per-anchor Diebold–Mariano tests of fBm against the dynamic-bundle mfBm at $\alpha = 0.05$, $h = 1, \ldots, 30$ and $d = 1, \ldots, 9$ — share of anchors at which DM rejects in favour of fBm under QLIKE (significant deterioration). Compact view (Panel B of Table 4).

$h\backslash d$	1	2	3	4	5	6	7	8	9
1	0.05	0.05	0.07	0.14	0.19	0.21	0.23	0.30	0.30
2	0.02	0.00	0.02	0.05	0.05	0.07	0.12	0.09	0.09
3	0.00	0.00	0.00	0.00	0.00	0.02	0.02	0.02	0.02
4	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00
5	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
25	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
27	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
28	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
30	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 12: Per-anchor Diebold–Mariano tests of fBm against the dynamic-bundle mfBm at $\alpha = 0.05$, $h = 1, \ldots, 30$ and $d = 1, \ldots, 9$ — share of anchors at which DM rejects in favour of fBm under MSFE (significant deterioration). Compact view (Panel C of Table 4).